

Appendix A1: Structured Literature Review Protocol

1. Purpose

This appendix records the structured literature review protocol used to define the review scope, search themes, screening criteria, extraction fields, synthesis approach and citation-validation rules.

2. PICOC Framing

Table A1.1: PICOC framing

PICOC element	Dissertation framing
Population	Official, publicly available SEC 10-K filings, with related financial long-document studies used as supporting context.
Intervention	Direct long-context prompting, where the full cleaned filing is supplied to the selected GPT-5.5 API model.
Comparison	RAG-based prompting, where retrieved chunks from the same cleaned filing corpus are supplied to the same model.
Outcomes	Claim-classification performance, qualitative error and evidence-use patterns, token use, telemetry-estimated model-call cost and latency.
Context	Financial long-document reasoning and evidence-grounded LLM evaluation under a Type 5 experimental design.

3. Source Families and Search Themes

Searches were organised around dissertation themes rather than open-ended keyword accumulation. Appendix A2 records the validated sources, final links and source-specific claim boundaries. The search themes are summarised below.

Table A1.2: Source families and search themes

Theme	Purpose / example search family
T1: Long-context LLMs	Long-context models, document-scale reasoning, large context windows and long-context benchmarks.
T2: Long-context limitations	Position sensitivity, lost-in-the-middle effects, effective context use and long-context degradation.
T3: RAG and retrieval-based prompting	RAG foundations, retrieval-mediated prompting, retrieval-augmented generation workflows and long-document RAG.
T4: RAG limitations	Retrieval miss, noisy retrieval, hard negatives, chunking, evidence fragmentation and RAG failure points.
T5: Financial-document reasoning	SEC 10-K filings, public-company financial QA, financial long-document reasoning and financial benchmark datasets.
T6: Evidence grounding and hallucination	Factuality, faithfulness, hallucination, grounded generation and evidence-based evaluation.
T7: Claim verification and inconsistency detection	Claim/evidence/decision framing, contradiction detection and financial-report conflict detection.
T8: Evaluation methods	Accuracy, precision, recall, F1-score, qualitative error analysis, reproducibility, cost and latency.
T9: Official source and implementation facts	SEC 10-K filing structure and official model/API capability, settings and pricing documentation.

4. Inclusion, Exclusion and Demotion Criteria

Table A1.3: Inclusion, exclusion and demotion criteria

Decision	Rule
Include	Sources that directly support the final scope: long-context LLMs, RAG, retrieval limitations, financial-document reasoning, SEC 10-K structure, evidence grounding, hallucination, claim verification, contradiction detection, evaluation metrics, cost, latency or experimental methodology.
Include with role restriction	Official SEC documentation and official model/API documentation support official facts only, such as filing structure, model capability, pricing or rate limits. They do not support academic performance claims.
Include older sources with temporal restriction	Pre-2020 sources may be retained for stable concepts such as classification metrics, experimental methodology, accounting/financial-disclosure context or regulatory background. They should not be used to support current LLM/RAG capability or performance claims.
Use preprints cautiously	arXiv and other preprints may be used for recent LLM/RAG work where peer-reviewed versions are unavailable. They must be labelled and not overstated.
Exclude or demote	Sources focused mainly on LoRA, domain-adapted model comparison, broad production finance automation, broad summarisation, risk-factor extraction as the primary task, narrative drift as a separate task, unrelated NLP tasks, unsupported blogs, unclear provenance or non-English material unless supervisor-approved.

5. Screening, Extraction and Recording

Candidate sources were screened by title/abstract and, where required, full text. Included sources were logged in Appendix A2 with source type, search theme, supported claim, support strength, pinpoint location and citation boundary. Only sources recorded in Appendix A2 were cited in the dissertation.

Table A1.4: Screening extraction fields

Extraction field	Purpose
Bibliographic identity	Authors, year, title, venue, DOI/arXiv/official URL and final reference form.
Source role	Academic literature, preprint, official documentation, professional guidance, implementation support or background source.
Method and task details	Dataset/domain, task, method, metric, experimental setting and whether the source addresses the same or an adjacent question.
Dissertation mapping	Chapter/section use, supported dissertation claim, source pinpoint, support strength and citation boundary.
Quality note	Limitations, overclaim risk, demotion/exclusion reason and link-check status.

6. Synthesis Method

The review used critical thematic synthesis rather than a paper-by-paper summary. Sources were grouped by dissertation theme, including long-context capability, long-context limitations, RAG-based prompting, RAG limitations, SEC 10-K reasoning, evidence grounding and evaluation methods. The table below shows how this synthesis was operationalised.

Table A1.5: Synthesis methods

Synthesis mode	Use in the dissertation
Taxonomic synthesis	Groups literature by concept and task: long-context capability and opportunity, long-context limitations, RAG baseline, RAG limitations, SEC 10-K reasoning, evidence grounding and evaluation methods.
Matrix synthesis	Uses Appendix A2 to compare source role, method, dataset, task, claim support, limitation and chapter mapping.
Template synthesis	Applies a consistent extraction template so sources can be compared fairly.
Method-aware synthesis	Explains not only what prior studies found, but how they studied the problem and whether their methods are sufficient for this dissertation gap.
Objective critique	Records uncertainty, limitations and transfer boundaries rather than assuming that long-context prompting or RAG is superior before the experiment.

7. Citation Validation Rules

Table A1.6: Citation validation rules

Rule	Requirement
Exact claim support	A citation must support the exact sentence or claim to which it is attached, not merely the general topic.
Pinpoint support	Appendix A2 records where support appears in the source, such as abstract, section, table, benchmark description, official form item or methodology section.
Boundary control	Appendix A2 records what each source does not prove, preventing benchmark, survey or official-document sources from being overused.
Clickable links	Final reference links and matrix source links use clickable resolver URLs where possible, such as DOI resolver links or full arXiv links.
No unused references	Every source in the final reference list must be used in the dissertation text or appendices, and every cited claim must map back to Appendix A2.